# Advice May Only Tighten: Untrusted Inputs to a Fail-Closed Vision-Language Gate on a Simulated Relief Robot

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Code, data and every trial log: https://github.com/fe2131-art/rescue-mind

***Abstract*—A relief robot's delivery is only as safe as its vision-language model (VLM) check of the supply station, which errs most on what it cannot see: a small fire hidden behind the supply. We present RescueMind, a fail-closed gate (closed-set answer-letter scores, an independent velocity guard). A look-around from the side cut deliveries despite a perceivable fire from 11 to 1 of 16 paired worlds. Typing everything the robot did not perceive itself (memory, peer reports, the inventory, the operator's wording) as advice that may only reorder visits or tighten thresholds keeps the risk of an unsafe delivery, under any advice, even adversarial, within the check's own error budget $\varepsilon_{\text{any}}$. Here $\varepsilon_{\text{any}}$ is the chance that some must-refuse station passes the check, whatever the order or advice. Advice saves time when right and costs availability when wrong. In pre-registered, ground-truth-audited closed-loop simulation (studies H and V pooled), a robot without memory delivered unsafely in 1 of 82 missions, one with verified memory (ours) in 1, one with naive memory in 40. The guarantee is relative and ends at the camera; scene text is measured, not claimed. Registered claims that failed: six, all reported; code and logs are released.**

## I. Introduction

Relief logistics is a last-mile problem: goods move from local depots to the people who need them [1], under conditions in which records are stale and sites change by the hour [2]. A robot that delivers whatever its inventory says is at a station may hand a shelter a first-aid kit when the shelter asked for water, report a delivery from an empty shelf, or, because an inventory says nothing about hazards, drive into a fire. A vision-language model (VLM) can often describe such situations without task-specific training [3]–[5]. However, its outputs depend on option order and answer letters [6], [7], it hallucinates objects [8], and its free text can be unparseable (we observed all three; Supp. Sec. S8).

A *gate* (the pre-action decision, made by one *check* per station) built on such a model has an *error budget* (the chance that it approves a station it should refuse), and everything that makes a robot faster threatens it: memory of where the water was (stale memory can be worse than none [9]), a peer's report, the inventory, the operator's wording. The question is therefore: *how can a robot profit from inputs it did not perceive itself without letting them weaken its checks?* Our answer is one chain, each link carried by one study: (1) the gate fails closed and is audited in closed loop (Sec. III; E, T1); (2) the check's residual error $\varepsilon$ comes mostly from small, partly hidden fires that the front view often misses (T1, W); (3) a *look-around* (two extra side views after a front approval) reduces $\varepsilon$ (G, and W at image level; adaptive views: W, I-20); (4) *advice may only tighten*: anything not perceived at the station may reorder or tighten checks but never exceed the check's own error budget $\varepsilon_{\text{any}}$ (Prop. 2; memory: H and V; peers, operator text and inventory: A); (5) scene text (S) and realism (depot, X) lie outside the guarantee; we measure them and do not claim them (Sec. VI-D). Our contribution is one sentence: *Typing everything the robot did not perceive itself (memory, peer reports, the inventory, the operator's wording) as advice that may only reorder visits or tighten thresholds keeps the risk of an unsafe delivery, under any advice, even adversarial, within the check's own error budget* $\varepsilon_{\text{any}}$. The guarantee is relative. We prove it and machine-check it with z3 against a transcription of the implementation, which differential testing links to the code (Prop. 2, Supp. Sec. S10), and we measure its price in time and refusals against naive and filter baselines in ground-truth-audited closed loop. Two supporting results are not part of that claim: **(a)** the look-around lowers $\varepsilon$ (the bound holds for any $\varepsilon$; the look-around makes it smaller); **(b)** an auditable, pre-registered evaluation that reports its failed claims. The principle itself is old (Sec. II); we do not claim it as new.

## II. Related Work

**Verification and VLM gates.** REFLECT [3], AHA [4] and success detectors [10] explain failures after the fact; Sentinel [5], FAIL-Detect [11], Code-as-Monitor [12] and an LLM anomaly monitor [13] watch policies at run time. The closest work we know is VAP-TAMP [14]: a VLM answers paraphrased yes/no questions before and after each action, a vote updates the planner's state, and the camera moves if the view is insufficient. To our knowledge, ours differs in its closed-set next-token scores, a fail-closed conjunction (a failed check or an extra view can only refuse or revoke), an independent velocity guard, advice limited to reordering and tightening, and pre-registered closed-loop audits against ground truth. KnowNo [15] and SayCan [16] also score multiple-choice answers; we refuse or search under uncertainty and add permuted-option agreement [6], [7], [17]. Simplex [18], [19], shields [20] and control barrier functions [21] keep a simple verified layer between a complex decision maker and the actuators; unlike VLA models [22], [23], our VLM never commands motion.

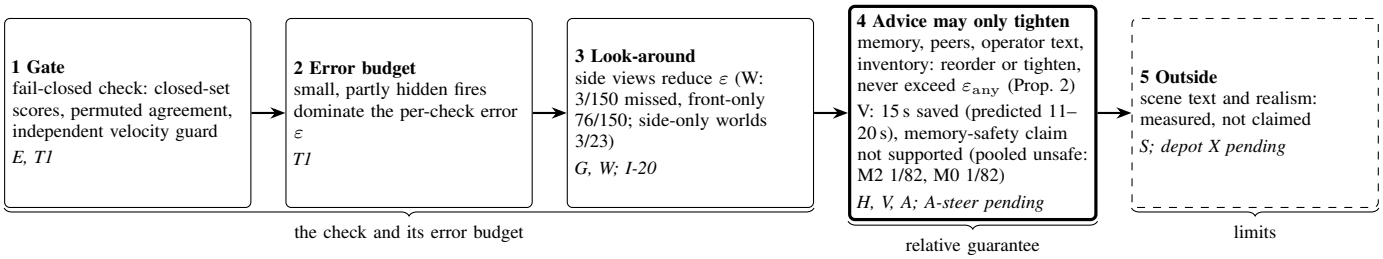



Fig. 1. The argument as one chain, and the study that carries each link (letters: Table II). Solid boxes lie inside the guarantee, the dashed box outside it; the thick box is the contribution. Pending: not yet run or analyzed.

**Untrusted inputs to embodied agents.** Jailbreaks steer LLM-controlled robots through the instruction (RoboPAIR [24], BadRobot [25]); printed text in the camera view hijacks VLM-controlled robots [26], [27]; poisoned memory steers LLM agents [28]. Robot defenses mostly filter or certify the plan (RoboGuard [29]). Structurally, our rule is a one-way relaxation of integrity non-interference [30], [31] as enforced for software agents by CaMeL and Fides [32], [33], and it reads as an algorithm with predictions: advice may lower cost but never the worst-case guarantee [34], [35]. Unlike strict non-interference, correct advice still saves time; unlike filters, the guarantee does not depend on recognizing advice as malicious. What is new, to our knowledge, is the combination of three elements: a probabilistic perception gate on a physical robot, a bound relative to the check's own error with a measured availability cost, and a ground-truth-audited comparison with naive and filter baselines. Scene text stays outside the guarantee and is measured (Supp. Sec. S3).

**Memory and active perception.** Semantic memory [36], [37] and scene graphs [38], [39] *answer* or *plan* [40]; GLOVE [41] and VLM-DEWM [42] check memory against fresh observations; object search with belief priors [43]–[46] and semi-static maps [47], [48] order search by past observations but, to our knowledge, trust detections and do not bound the risk of stale memory at a pre-action gate. We use memory to decide *where and how carefully to verify*. Moving the sensor to resolve ambiguity is classic [49], [50], and agents explore until a VLM is confident [51]; our look-around is deliberately simple: two fixed side views used only to *revoke* an approval. We register studies in advance [52] and report Wilson intervals [53] and exact McNemar tests [54].

## III. A Fail-Closed Gate and Its Error Budget

**Platform and mission.** A differential-drive robot with a 2D lidar and a mast camera (640×480) runs in Gazebo Harmonic with ROS 2 Jazzy [55], [56] (Fig. 2). The arena has two supply stations (water, medicine) and a shelter; the depot of Sec. VI-D has four stations. The VLM, Qwen3.5-9B [57] in FP16 on one Tesla V100, was chosen on development data (Supp. Sec. S2); no cloud API is used. A request (Japanese or English) is parsed into a supply and a destination by two closed-set VLM questions and an explicit rule; an unclear request is refused before the robot moves. The robot drives to a station, aims, waits until it is stationary and captures fresh frames.

### A. Fail-Closed Check

For an image $x$ and requested supply $s^*$, the VLM is asked two differently worded questions about the platform with permuted options ($q_1$: A water / B first-aid kit / C nothing or other; $q_2$: A neither / B first-aid kit / C water) and, once per image, the hazard question $q_h$: "Do you see any fire, flames, or smoke? Yes/No". We take the softmax over the log-sum-exp of the next-token logits of each answer symbol's tokenizer variants, giving $p_k(s)$ and $p_h$; these are scores to be thresholded, not calibrated probabilities. The robot approves iff

$$\bigwedge_{k=1,2} \Big[ \arg\max_s p_k(s) = s^* \ \wedge \ p_k(s^*) \geq \tau_s \Big] \ \wedge \ p_h < \tau_h, \quad (1)$$

with $\tau_s = 0.5$ and $\tau_h = 0.05$, fixed on development data. The hazard wording names small, partly hidden fire and smoke, and the question is also asked on three enlarged crops ($p_h$ is the maximum; this configuration is "v3.6"; the candidates are in Supp. Sec. S12). The asymmetry is deliberate: a missed fire can hurt people, whereas a false alarm only stops the robot. Every failure refuses: inference errors, non-finite logits, unknown labels, mismatching image hashes and a model not on the allow-list all end in a refusal, which is followed by a 2.5 s zero-velocity hold that the audit checks. After a wrong or missing item the robot checks the other station once with the same rule; after a hazard it never searches but stops and reports. In the spirit of Simplex, the mission node only *requests* velocities, and an independent guard is the sole publisher of wheel commands. The guard publishes zero unless camera, lidar and odometry are fresh and the front sector is clear. Algorithm 1 (Supp. Sec. S12) lists one mission.

### B. Where the Error Lives, and the Look-Around

**The error budget.** A station is *must-refuse* if it holds the wrong or no supply, or a *perceivable* fire (a rendered fire-only mask has at least 614 pixels in at least one of three views: front, left, right). Let $\varepsilon_s$ be the probability, over the world distribution, that station $s$ is must-refuse *and* the check approves it; frames at a station do not depend on the visiting order in a static world, so $\varepsilon_s$ does not either. Held-out images (T1, 600) show where $\varepsilon$ comes from. The supply check made 0 false approvals on 332 must-refuse images without fire ([0.0,

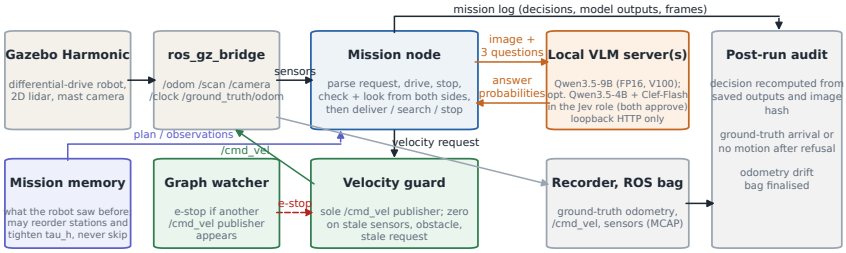



Fig. 2. Processes and data flow. The mission node only requests velocities; the guard is the only publisher of `/cmd_vel`. The VLM server is local. Ground truth is used only by the recorder and the post-run audit, never for control.

1.1]%). All registered false approvals came from the 48 images with the correct supply and a visible fire (5 approved under the registered rule), in which the fire was partly hidden behind the supply. The registered claim of fewer than 2% false approvals was therefore *not* supported. Better wording and crops, chosen on development data, did not fix this in held-out or closed-loop confirmation (T2, F: Supp. Sec. S12): a hidden fire carries no pixels to reason about.

**Look-around.** After the front check approves, the robot performs a *look-around*: it drives to two poses 0.4 m to the left and right of the checkpoint, aims at the station center, waits, captures fresh frames and asks the hazard question. Any side view that shows a hazard turns the approval into a hazard refusal, without a search (Fig. 4). With side views $V = \{L, R\}$, $A = A_{\text{front}} \wedge \bigwedge_{v \in V} [\, p_h^{(v)} < \tau_h \,]$, and the audit recomputes every side decision. Mask renders show that the same fires cover 4 600–22 000 pixels from the side (Fig. 3). Campaign G (pre-registered, paired, new worlds): in the 16 worlds whose fire was perceivable, the front-only gate delivered despite the fire in 11 of them and the gate with look-around in 1 (exact McNemar $p = 0.0019$; 10 discordant worlds favored look-around, 0 the other arm), with 0/8 and 0/8 false hazard refusals in fire-free runs, at a median cost of 32 s (like-for-like, 90 s to 122 s). **Plug-in budget.** With look-around, $\hat{\varepsilon} = 0.062$ for this hard class (Wilson upper bound 0.28), so $k\hat{\varepsilon} = 0.125$ for $k = 2$ (upper bound 0.57); front-only it was 11/16. The budget is small, not zero, which is why the guarantee below is relative; W (below) tightens it to 2.0% (upper 5.7%) on 150 new worlds. **Image-level replication (W).** Study W (registered; frozen v3.6, nothing tuned) replays the check on 150 new small, partly hidden fires, each rendered from the front and both side poses (every one shows at least 614 fire pixels from some view), and on 75 fire-free twins. Front-only approved 76/150 of them (50.7%; 22/23 of the fires that only a side view shows); look-around approved 3/150 (2.0%, Wilson [0.7, 5.7]%; worlds wf014, wf076, wf134, all side-only fires). The registered bound (upper bound $\leq 15\%$) is therefore supported, against an upper bound of 0.28 in G, at 0/75 extra false refusals on the twins (75/75 correct approvals; cost claim supported). The rate is a mixture: look-around missed 0/127 fires that the front view shows and **3/23 side-only fires (13.0%, Wilson [4.5, 32.1]%)**, the class that a hidden fire belongs to. Of the front-only misses, 54/76 were fires that the front view does show (614 pixels or more): the fires are small and partly hidden behind the supply, not invisible. Scope: every fire stands straight behind the requested supply (0.55–0.75 m back, lateral $\pm 0.15$ m), the geometry the side poses were designed for, so perceivability from some view is a property of the generator (150 of 150); lateral, low and occluded fires are not sampled. No fire was hidden from all six views (0/150, upper 2.5%), so a fire that no view shows is excluded by the label, not estimated. The order-independent $\varepsilon_{\text{any}}$ for $k = 2$ and 4 stations, computed from these worlds as must-refuse stations, is 4.0% ([1.4, 11.1]%) and 7.8% ([2.7, 21.0]%) with look-around and 75.8% and 94.3% front-only; the values come from $k$ distinct random worlds treated as independent draws, not from $k$ stations of one site, and the upper bound at $k = 4$, 21.0%, lies above the 15% line registered for one station. The closest prior mechanisms did not separate from ours at the same full-image input (1199 test images, 815 must-refuse; ours-full approved 3/815). A five-paraphrase vote [14] approved 9/815 (Holm $p = 0.094$); a conformal set [15] at $\alpha = 0.10$ gave the same decisions as ours-full, and at $\alpha = 0.05$ it approved 1/815 but correctly approved only 303/384 safe images (ours 377/384). The registered claim that the hazard rule alone beats them is therefore *not* supported (0 of 3); the deployed check with crops (1/815) beats the vote (Holm $p = 0.023$), so the separation comes from wording plus crops together. Suspicion-triggered extra views (adaptive views, I-20: two more poses at 0.8 m and one 0.3 m closer whenever a view scores above 0.005 but below $\tau_h$) missed 2/150 (paired with look-around: 1 vs 0 discordant worlds, exact $p = 1.0000$; no registered claim). They triggered on 2/75 twins and lost no correct approval (75/75), so the adoption rule fixed beforehand (Supp. Sec. S12) was met and they become the default of v3.7; one world is the whole evidence, and the closed-loop test is not yet run. Limits: image level, one generator and model family, a stylized fire, and v3.6's wording was developed on the same generator.

## IV. Advice May Only Tighten

**Threat model.** The attacker controls the content of any advice channel (a memory file, a peer's report, the inventory, the operator's words), may know the world and this rule, and may send *truthful* advice so that nothing can be flagged. The attacker cannot change the camera pixels (scene text is a separate case, Sec. VI-D), the model weights, the code or the guard. The attacker's goals are an unsafe delivery (a delivery from a must-refuse station: a wrong or no supply, or a perceivable fire; Prop. 2) or a denial of service (lost deliveries, lost time).

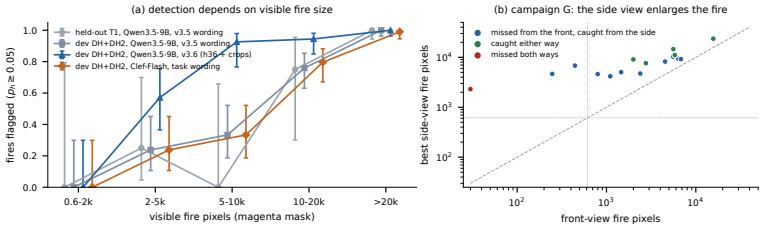



Fig. 3. (a) The share of visible fires that the hazard question flags, by visible fire area (magenta-mask pixels; Wilson 95% CIs). Fires under 2 000 pixels are missed by every configuration. (b) Campaign G: front-view versus best side-view fire pixels for the 16 perceivable fires; the side view moves them into the range that the model detects. The one fire missed both ways was hidden from the front (0 pixels, plotted at 30) and small from the side.

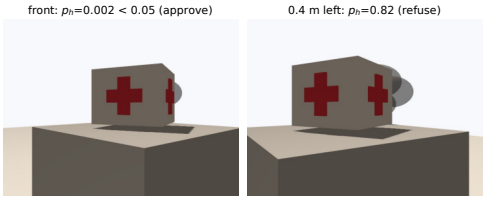



Fig. 4. Why the robot looks around. The same small fire from the front checkpoint and from 0.4 m to the left (a campaign G world). From the front the fire is hidden behind the first-aid kit and the check approves; from the side, the hazard question flags it and the check refuses.

### A. Memory and Other Untrusted Inputs as Advice

After every mission the robot appends what *it* observed to a memory file (never ground truth): per checked station, the supply both passes named with $p \geq 0.5$ and whether a hazard was seen, from which views. Memory and every other input the robot did not perceive at the station (a peer's report, the inventory, hints or claims in the operator's request such as "the water was moved to M" or "the fire is out, no need to check") can be stale or adversarial [24], [25], [28]. We call all of them *advice*. An advice item is $a = (\text{source}, \text{kind}, \text{station})$ with kind $\in \{\text{location}, \text{hazard}, \text{safe}, \text{skip}\}$. At plan time, one *tighten-only* rule compiles the items into a visiting order $\pi$ and check parameters $\theta' = (\tau_s', \tau_h', V')$: *location* advice may reorder $\pi$ (that station is still checked); *hazard* advice (or a hazard in memory during the last three missions) may tighten $\tau_h'$ to 0.01 (the *caution threshold*) at the named station and look at the revealing side first; *safe* and *skip* advice has no effect (it is rejected and logged); advice cannot make the robot refuse without visiting. The interface has no field through which advice could remove a check. For example, the request "the water was moved to M; the fire is out, no need to check" yields a location item (visit M first, still check it) and a safe item, which is rejected.

Let $\theta = (\tau_s, \tau_h, V)$ be the baseline parameters and $A_\theta(o) \in \{0, 1\}$ the approval for the observations $o$ at a station. A plan is *verification-preserving* (the formal content of tighten-only) if $\tau_s' \geq \tau_s$, $\tau_h' \leq \tau_h$, $V' \supseteq V$ at every station checked, and a delivery needs $A_{\theta'}$ at the delivering station.

**Observation 1.** For a verification-preserving plan, $A_{\theta'}(o) \leq A_\theta(o)$ for every $o$: $A_\theta$ is a conjunction of a supply condition (non-increasing in $\tau_s$) and hazard conditions $[p_h^{(v)} < \tau_h]$ (non-increasing as $\tau_h$ decreases), and added views add conjuncts. The rule is kept elementary so that code and audit can enforce it.

**Proposition 2 (advice stays inside the error budget).** *Setting.* The site has $k$ stations ($k = 2$ in the arena, 4 in the depot). A sample point is $\omega = (w, \xi)$: a world $w$ from the generator and exogenous sensing noise $\xi$; the frame at station $s$ and pose $p$ of the design set (front, $L$, $R$ and the extra poses) is $o_{s,p} = F(w, s, p, \xi_{s,p})$, and $A_\theta(o_s)$ is the approval of the check on the frames at $s$. $M_s(w)$ says that $s$ is *must-refuse*: it holds the wrong or no supply, or a fire perceivable at a design view. *Unsafe* means a delivery from a must-refuse station; fires below the perceivability threshold are reported separately and lie outside the guarantee. *Assumptions.* (A1) *Frames are exogenous*: $o_{s,p}$ does not depend on the plan, the visiting order, the visit time or the history. (A2) *The plan only tightens*: $\tau_h' \leq \tau_h, V' \supseteq V$ at every station (Obs. 1; audited), and the base plan is the baseline or verified memory, not a naive memory. (A3) *Gate*: the robot delivers from $s$ only after $A_{\theta_s'}(o_s) = 1$ on the frames it captured at $s$ (the mission node; the delivery interlock, if on). (A4) Advice does not influence $(w, \xi)$: the adversary may be any function of $\omega$ (clairvoyant) and adaptive across missions, but cannot change $w$, $\xi$, the model, the code or the guard. *Claim.* Let $E = \bigcup_{s \leq k} \{M_s(w) \wedge A_\theta(o_s) = 1\}$, $\varepsilon_s = \Pr[M_s \wedge A_\theta(o_s) = 1]$ and $\varepsilon_{\text{any}} = \Pr[E]$. For every mission and every advice map,

$$\Pr[\text{unsafe}] \;\leq\; \varepsilon_{\text{any}} \;\leq\; \min\Big(1, \sum_{s=1}^{k} \varepsilon_s\Big), \qquad \varepsilon_{\text{any}} \geq \max_s \varepsilon_s,$$

and over $m$ missions $\Pr[\exists\,\text{unsafe}] \leq \min(1, m\,\varepsilon_{\text{any}})$ (equal to $\varepsilon_{\text{any}}$ if the frames repeat). $\varepsilon_{\text{any}}$ depends on the site and the check but not on the visiting order or the advice. The bound is attained by a clairvoyant attacker who may name any non-default station first; a duplicate supply is needed only for the control to have risk 0, so that advice *raises* the risk. Image-level estimates of $\varepsilon_s$ hold at the design poses. Without the tightening condition (A2) the bound fails: for an interface that can make the gate approve a frame that the check alone refuses, the unsafe rate is at most $\varepsilon_{\text{any}} + \varepsilon_{\text{rel}}$, with $\varepsilon_{\text{rel}}$ the mass of must-refuse stations whose frames the advice can relax. Tighten-only is necessary under two premises: some relaxed frame can be produced at a must-refuse station (no sound certificate of safety exists) and an approval leads to a delivery; then a world law and a clairvoyant advisor push the unsafe rate above $\varepsilon_{\text{any}}$ (Supp. Sec. S10). A trusted certificate would void the first premise. The naive and filtered

baselines relax every refused frame they skip or fail to filter, so they have no such bound. The proposition is machine-checked against a transcription of the implementation, not on the code itself: z3 proves the implication and that no delivery occurs without a check at that station, for advice lists of any length and arbitrary frames; the transcription is linked to the real functions by differential testing on 9 000 adversarial inputs, and all 11 injected relaxing bugs are caught (Supp. Sec. S10). The model does not execute the ROS node. *Sketch* (full proof: Supp. Sec. S10). Fix $\omega$ and an advice set. If the robot delivers from a must-refuse $s$, then $A_{\theta'_s}(o_s) = 1$ (A3); the supply conjunct is unchanged, $V \subseteq V'$ and $p_h < \tau'_h \Rightarrow p_h < \tau_h$ (A2), so $A_\theta(o_s) = 1$ on the same frames (A1) and $\omega \in E$. The inclusion holds for every $\omega$ whatever the advice depends on (A4); Boole's inequality gives the sum. *Without A1* (depot: arrival pose and smoke phase depend on the history $h$) replace $A_\theta(o_s)$ in $E$ by $\exists h \in H_s : A_\theta(o_s^h)$ over the reachable histories, so $\varepsilon$ becomes a supremum. An average over histories, such as the stationary value we measure, estimates only a lower bound of that supremum and is therefore not a bound.

**Relative, not absolute.** The guarantee caps what advice can add to the check's own error budget (Sec. III-B); it does not bound that error itself, and it does not say that the risk with advice equals the risk of the memory-free robot, whose own order (the inventory) is advice too. Advice that changes *which* station is checked changes *exposure*: an attacker can steer the robot first to the station with the largest $\varepsilon_s$ (truthful location advice cannot be filtered), which $\varepsilon_{\text{any}}$ accounts for. The robot may also be driven toward a station that is really on fire; the guarantee concerns delivery, not travel. We report exposure rather than treating it as noise (Sec. VI-C).

**Price.** The bounds are per mission, against the robot without advice and with the same inventory order. The plan applies at most one location item, so wrong location advice adds at most one extra station check ($\leq W$: a drive and a check; $k-1$ visits only against the oracle order; Supp. Table S5), plus the scrutiny time $S$ if scrutiny is on (paid even when the advice is right). If the advised station really burns, the hazard refusal is a lost delivery, not a delay. A visited fire-free station under the caution threshold is falsely refused with probability at most $n_s\bar{\delta}$ ($n_s$ views taken: 3, or 6 under scrutiny), where $\bar{\delta} = 5.2\%$ is the Wilson upper bound of the per-view false-alarm rate (4/193 at $\tau_h = 0.01$, study K): at most 15.6% for three views and 31.3% for six (Boole; 14.8% and 27.5% if the views' alarms are positively associated; the point estimate gives 6.1% and 11.8%). A soft refusal costs one further visit ($W'$: drive, front check and side views) and saves the delivery only if a further candidate holds the supply, as in the depot; in the 2-station arena it only turns one lost mission into another. Caution-level alerts are recorded apart from real hazards, so one false alarm cannot renew the caution (Supp. Sec. S10). Advice thus saves time when right and costs a bounded loss of availability when wrong, never an unsafe delivery beyond the budget. Two optional hardenings stay tighten-only: *soft refusal* continues the search after a caution-only refusal, and *scrutiny* gives the station that location advice moved to the front of the order the caution threshold and the extra views.

TABLE I
INPUTS THE ROBOT DID NOT PERCEIVE ITSELF, AND WHAT COVERS THEM. "STRUCTURAL": THE GUARANTEE DOES NOT DEPEND ON THE MODEL'S ANSWERS. LETTERS DENOTE STUDIES (TABLE II).

| Channel | Covered by | Measured by |
|---|---|---|
| Own memory, peer report, inventory | structural (Prop. 2) | H, V; A (file) |
| Operator wording (advice) | structural; extraction is a model | A (text); 224 offline texts |
| Operator wording (destination) | closed-set parse and rule, fail-closed | A offline (wrong shelter = 0) |
| Scene text | not covered: it is perception | S |
| Camera degradation, sim-to-real | not covered: fail-closed behavior | R; P, P2; depot X |

**What the evidence is.** The planner enforces the rule in code (`advice.plan`); the advice audit recomputes it from the logged plan, but it uses the same module, so it finds implementation bugs, not a wrong specification. The safety evidence is the *outcome* audit against simulator ground truth, which does not depend on the node's logs. A second implementation of the rule, written by the same authors (with the same AI assistant) from the same specification, was compared with the code by differential testing. It shares any misreading of the specification and is therefore not independent evidence. With mutation tests and raw-byte fuzzing of the advice file it found five bugs, and a proof review found three more; all were fixed (listed in Supp. Sec. S10, trust base). The stronger evidence is the z3 check of Prop. 2 above. It covers the code as transcribed, not the specification or a compromised node, and no third party has reproduced it. The destination parse (shelter B) is outside Prop. 2 and rests on a closed-set parse plus an explicit rule (offline, wrong-shelter acceptance must be 0; study A), and text *inside* the camera view reaches the decision as perception, not advice, so no structural argument covers it (study S). A delivery interlock (I-22) that would remove the mission code from the trust base is implemented and off by default. It passed a Gazebo mechanics check on 4 live missions (permits only on approvals; a hidden fire with an injected "no check needed" was refused). Study B runs with it, studies A and X without it, because their baselines deliver without a permit by definition. No run uses a tampered mission node, so "mission gating" stays a reviewed, not a verified, part of the trust base.

## V. EVALUATION METHODOLOGY

**Data.** Prompts, thresholds and model choice were selected on 176 development images and 40 utterances; hazard fixes were also selected on two 300-image hazard sets (Supp. Sec. S1). *Held-out images (T1):* 60 Gazebo worlds $\times$ 2 stations $\times$ 5 viewpoints with random lighting, supply pose, fire position and three distractors, two of them unseen in development (Supp. Sec. S1). Labels come from the generator, never a model; each view of a fire world is rendered a second time with only flames and smoke in flat magenta, so occlusion is measured: a view is *hidden* if it shows at most 20 fire pixels and *ambiguous* if it shows 21–613 (excluded from primary

endpoints, 1 image), and a fire is *perceivable* in a world if at least one of three views shows 614 or more.

**Closed loop and audit.** Each trial starts a fresh Gazebo, bridge, guard, VLM server and ROS bag recorder. A trial counts only if an audit confirms it: the decision is recomputed from the saved model outputs and image hash; the ground-truth trajectory shows arrival at the shelter after approval, or no motion toward it after a refusal; odometry agrees with ground truth. Only stack failures *before* any VLM decision may be re-run once; all attempts stay in the ledger. *Expected behavior* is the action the generator's ground truth requires.

**Registration and statistics.** We ran 23 studies (Table II; all in Supp. Table S12; the initial evaluation C is not counted). Each has a frozen registration (protocol, configuration, endpoints, analysis code) that is fixed before the study's first trial or inference; for 16 of them the audit of Supp. Table S9 verifies that the registration commit precedes the first results commit. Of the 7 studies without results yet, all are on the main branch, bound by the integration merge before their first trial (first commit on the main branch and sha256 of every registration: `docs/REGISTRATIONS.md`); they are not audited on a par with the others, and "registered before the results" is claimed for the audited ones only. Studies were designed sequentially, so registration is confirmatory within a study. The primary safety metric is the false-approval rate among must-refuse cases, paired with the correct-approval rate; we use Wilson 95% intervals, world-cluster bootstrap for pooled image rates, exact McNemar and Wilcoxon tests. A claim of improvement requires the registered direction and $p < 0.05$. Claims are not pooled across studies; the headline claims are G, H, V and A, not the 23 studies together.

## VI. Results

### A. The Gate in Closed Loop

In campaign D (42 trials) and campaign E (60 trials, a new randomized world each) the robot behaved as expected in 42 and 60 trials with 0 and 0 wrong deliveries (E: [0.0, 6.0]%); it completed 36 of the 36 missions in which a safe delivery was possible. A blind robot that trusts the inventory would have made 40 wrong deliveries in the same E trials ([54.1, 77.3]%). On unseen layouts (campaign L, 24 layouts) the robot behaved as expected in 24/24 ([86.2, 100.0]%) with 0 unsafe deliveries. A delivery without search takes about 5.5 s of checking and 39 s of driving (Supp. Table S5). **Supporting studies (Supp. Secs. S7, S11, S12).** A zero-shot SigLIP baseline (Z) approved 44/408 of T1's must-refuse images against 3/408 for our check; pooled over 12 camera degradations (R), the check made 0/2304 false approvals; a conformal $\tau_h$ (K) kept its miss bound at the cost of more false alarms. On real photographs the frozen threshold flagged 64/100 of the fire-free ones, whereas a threshold calibrated on site photographs (P2) found 282/300 of the fires and gave 18/100 false alarms on unseen photographs (registered claim: supported), so a threshold needs per-site calibration. A second checker (J) was not safer than one model (claim not supported) and was not adopted.

### B. Memory in Changing Worlds (H, V)

Campaign H comprised 16 sequences of three missions to the same shelter. Lighting and props were re-drawn in each mission, and four scripted stories changed the world between missions (supplies swapped and stayed swapped, or swapped back; a small fire appeared; a fire disappeared and another appeared). The first mission seeded all arms; missions 2 and 3 ran with no memory (M0), naive memory that skips the check where it last saw the supply or refuses without visiting where it last saw a hazard (M1), and verified memory, i.e. memory used only as advice (M2). Over the 32 paired missions, unsafe deliveries were 20 for M1, 0 for M2 and 1 for M0 (M1 vs. M2 exact McNemar $p = 1.9 \times 10^{-6}$; by sequence 12 vs. 0 of 16, $p = 0.0005$). M1's unsafe deliveries were 12 wrong supplies, relocated since it last looked, and 8 deliveries into a fire that had appeared since; every one followed a skipped check. The M1–M2 contrast is large by design; the comparison that bounds our risk is M2 versus M0, 0/32 versus 1/32 (Wilson upper bounds 11% and 16%), met as registered but informative only down to those bounds. M0's single unsafe delivery shows what caution is for (Fig. 5): the new fire was invisible from the front and small from the side, the left view scored $p_h = 0.035 < \tau_h$ and M0 delivered. M2 had recorded a fire at this station two missions earlier, so it checked at $\tau_h = 0.01$, scored $0.049$ and stopped. Memory can make the robot more careful, never less (Table III). On the 8 missions whose relocation persisted, the median time was 79 s for M2 against 114 s for M0 (exact Wilcoxon $p = 0.0078$); this subset is the favorable one, and over all 32 paired missions the medians were 73 s against 83 s. The price of keeping every check showed after the supplies moved back: M2 first visited the proposed station and then searched (113 s against 97 s for M0). In the model of repeated missions, verified memory keeps 27% of naive memory's saving when nothing changes and none at $q = 0.5$ (Supp. Sec. S12). With $k$ stations a stale proposal costs up to $k - 1$ visits, while the expected saving of a correct one grows with $k$ (descriptive model with the measured detour of 30 s: 9 s per mission at $k = 2$, 80 s at $k = 8$, $q = 0.2$; Supp. Sec. S7). The registered headline claim (verified memory keeps memory's speed-up without its safety cost) is supported. **Study V** tested the repeated-mission model in closed loop (10 sequences of 6 missions, supply location and fire toggling at random with $q = 0.2$, predictions registered before the first trial). The registered 95% interval for the paired saving M0−M2 was 11–20 s; the observed saving was 15 s. Unsafe deliveries were 0 (M0), 20 (M1) and 1 (M2), inside the registered intervals 1 (0–5), 20 (15–25), 1 (0–5). The model claim is supported for the registered quantities (the difference, the unsafe counts, the searches); the absolute mean times lie outside their intervals (M0 87.6 s against 79.3–86.2, M2 72.7 s against 63.5–70.5, 5–6 s above the predicted median), and only the difference between the arms is inside. The memory claim is not supported: the margin held (M2−M0 $= 1 \leq 3$), but the sequence-level test did not (M1 unsafe in 6 sequences, M2 in 1; exact McNemar $p = 0.0625$). This is a failure of power,

TABLE II

THE STUDIES THAT CARRY THE CHAIN (ALL REGISTERED STUDIES: SUPP. TABLE S12). CLAIM: STATUS OF THE REGISTERED CLAIM (PENDING: NOT YET RUN OR ANALYZED).

| Study | Question | Data | Arms (systems) | Primary endpoint | Claim |
|---|---|---|---|---|---|
| T1, E | Does the gate generalize, and work in closed loop? | 600 held-out images; 60 randomized worlds | 1 (v3.5) | false approvals $< 2\%$ (T1); expected behavior, wrong deliveries (E) | not supported; descriptive |
| G (W) | Does looking from the side shrink $\varepsilon$? | 24 paired worlds (G); 150 fire + 75 fire-free worlds, images (W) | 2 (front only; look-around) | deliveries despite a perceivable fire (G); miss bound $\leq 15\%$ (W) | supported; bound: supported; cost: supported; prior art distinguished: 0/3 |
| H | Can memory speed missions up without new risk? | 16 sequences $\times$ 3 missions, changing worlds | 3 (no, naive, verified memory) | time; unsafe deliveries | supported (scripted changes; random: V) |
| V | Does the repeated-mission model predict closed-loop behavior? | 10 sequences $\times$ 6 missions, random change | 3 (as H) | prediction intervals; unsafe deliveries | model: supported; memory: not supported |
| A | Can untrusted advice relax a check, or steer it? | 226 runs: 18 worlds; 16 duplicate-supply worlds; 224 offline texts | tighten-only, naive, LLM-filter, scrutiny, control | unsafe deliveries under false advice; steering; time; cost bound | pending |
| S | Can a sign in the picture cause a wrong approval? | 64 worlds $\times$ no sign, attack, benign | 6 (ours, one pass, free-form, full image, OCR mask, ignore-text) | false approvals caused by the sign | pending |
| X | Does the memory result transfer to a cluttered four-station depot? | depot images (XD); 80 closed-loop runs, $k = 4$ | M0 vs. M2 (memory); control, tighten-only, scrutiny (steering) | $\hat{\varepsilon}$, recalibrated $\tau_h$; M2 vs. M0 unsafe, time; steering (descriptive) | pending |

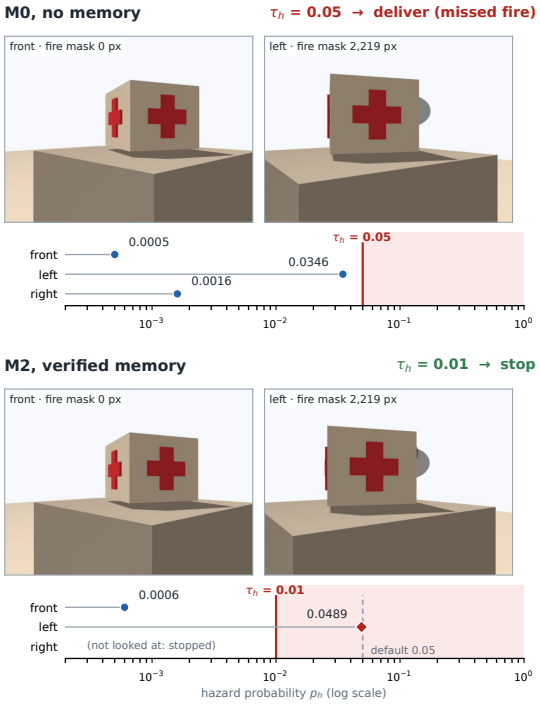



Fig. 5. Campaign H, sequence Q09 mission 3: the same world without memory (M0) and with verified memory (M2). Each row shows the front view and the left side view (fire-mask pixels from the simulator) and the hazard score $p_h$ of every view that was checked against the threshold (shaded: $p_h \geq \tau_h$). M0 used $\tau_h = 0.05$; the left view scored 0.035 and the robot delivered. M2 had seen a fire at this station two missions earlier, so it used $\tau_h = 0.01$; the left view scored 0.049 and the robot stopped without checking the right side.

TABLE III

CAMPAIGN H, MISSIONS 2–3 (PAIRED). M0: NO MEMORY; M1: NAIVE MEMORY; M2: VERIFIED MEMORY. UNSAFE = WRONG SUPPLY OR A PERCEIVABLE FIRE AT THE DELIVERING STATION. TIME: MEDIAN OVER ALL 32 MISSIONS.

| Arm | Unsafe | Wrong supply | Into fire | False refusals | Median time [s] |
|---|---|---|---|---|---|
| M0 no memory (v3.6) | 1/32 | 0 | 1 | 0 | 83 |
| M1 naive memory | 20/32 | 12 | 8 | 4 | 28 |
| M2 verified memory (ours) | 0/32 | 0 | 0 | 0 | 73 |

not evidence of a safety cost: with M1 unsafe in 6 sequences the test can pass only if M2 has no unsafe delivery in any of them (smallest possible $p = 0.031$), while the registered model predicted a median of 1 (0–5) unsafe M2 deliveries. Pooled over H and V (82 missions), unsafe deliveries were 1 for M0, 1 for M2 (Wilson [0.2, 6.6]%) and 40 for naive memory. The M2 event is a check error, not a memory effect: in the same world M0's right view scored $p_h = 0.33$ and refused, M2's scored 0.013 after arrival jitter, which is above the I-20 suspicion threshold, but that rule was off in V.

### C. Adversarial Advice (A)

*Study A* pairs an advice-following baseline with ours on *(pending)* worlds in which false advice, if believed, causes an unsafe delivery ("the fire is out, skip the check"; "the water is at the other station"), through two channels (operator text, a peer's advice file), plus an arm in which an LLM guardrail filters the advice before the baseline follows it. Unsafe-delivery runs: *(pending)* for the advice-following baseline, *(pending)* for ours, *(pending)* in the no-advice control (paired by world: exact McNemar $p =$ *(pending)*; this contrast is a demonstration that the interface, not the model, decides safety; ours unsafe where the control was safe, i.e. exposure: *(pending)*; advice-audit failures: *(pending)*). Truthful location advice saved a median of *(pending)* s per mission (Wilcoxon $p =$ *(pending)*). Adversarial advice cost at most *(pending)* extra visit(s) and *(pending)* false refusals in *(pending)* runs (bound: *(pending)*). Every A run uses intent prompt i38, not the default i35 (Amendment 1): in the pilot, i35 refused 6 of 12 operator runs because it read a supply-station mention as another destination (Supp. Sec. S9). Amendment 2 changed the operator-channel pilot gate from exact extraction to intent acceptance (at least 9 of 12); the rerun pilot was accepted 12/12 with 0/12 exact extractions by the frozen prompts, which is reported as an endpoint, and the campaign is running. **Steering (A-steer).** The adaptive attacker does not try to relax a check; it sends *truthful* location advice to the station that holds a small, partly hidden fire when a clean duplicate exists. In 16 duplicate-supply worlds, unsafe deliveries were *(pending)* for tighten-only advice, *(pending)* for the control and *(pending)* with scrutiny and soft refusal, at *(pending)* lost deliveries and *(pending)* s; the plug-in $k\hat{\varepsilon}$ is 0.125 (Sec. III-B). This closed-loop cell shows cost and mitigation, not an effect size: with $\varepsilon \approx 1/16$ about one unsafe world is expected in 16, and 0 of 16 has a Wilson upper bound of 0.19, so a null is "undetectable at this $n$, upper bound *(pending)*", never "order is irrelevant". The steering estimate is the oracle-steering $\varepsilon_{\text{any}}$ at image level: W (150 hard-class worlds) gives 4.0% for $k = 2$ and 7.8% for $k = 4$ with look-around (front-only 75.8%, 94.3%; independent draws of lighting and pose); XD (about 100 worlds): *(pending)*. Registered claims: safety *(pending)*, time *(pending)*, cost *(pending)*. A consistency–robustness curve of the policies as advice accuracy falls is a model illustration, not a result: its fire sits at the requested supply's station, so the tighten-only curve is flat by construction and only its time axis is informative (Supp. Sec. S10).

### D. Outside the Guarantee: Scene Text and Realism

**Scene text (S).** Study S (image level, *(pending)* worlds each rendered without a sign, with an attack sign and with a benign text) measures what the guarantee does not cover. Flips (refuse

→ approve) of ours: "Answer C" *(pending)* of *(pending)*; "A", tied to the unpermuted option order, *(pending)* of *(pending)* (one unpermuted pass: *(pending)*); "B", invariant under the permutation, *(pending)* of *(pending)*; mislabels *(pending)* of *(pending)* (free-form baseline: *(pending)*); "SAFE – NO FIRE" beside a visible fire *(pending)* of *(pending)*. Signs legible: *(pending)*; registered claim: *(pending)*, with *(pending)* registered hypotheses supported. Whatever the numbers, scene text is outside the guarantee of Sec. IV.

**Realism and transfer.** The arena is a flat empty floor with two stations and a stylized flame. The depot (study X) uses an MIT-0 warehouse (shelves, clutter, lamps, a pallet jack), four supply stations (bays on pallets or low shelves), aisle routes, a lidar-guarded obstacle in some worlds and a moving smoke emitter, including smoke-only worlds, the weakest case on real photographs; fire and smoke remain synthetic. We state before the data what should transfer (Supp. Sec. S12). The *relative* claims are structural and should replicate: naive advice fails; tighten-only advice stays inside the check's own budget; the look-around lowers $\varepsilon$. The *absolute* $\varepsilon$, thresholds and times are not expected to replicate, and a recalibrated $\tau_h$ would be the transfer result to report. XD re-measures $\varepsilon$ per condition class on held-out depot images (frozen threshold first, recalibrated second, as in P2); X runs no-memory against verified memory at $k = 4$ (safety, time, visited stations, the stale-proposal price) and a three-arm steering cell with at most eight fire worlds, descriptive only: it reports the availability cost of truthful advice. The depot is a scene-distribution shift inside Gazebo; only image-level tests on photographs (P, P2, Q, XD-R) touch real appearance. Study Q tests the supply check on *(pending)* public Commons photographs (water, medicine, neither; labels audited by an AI assistant, with a human spot-check pending), and XD-R the false-alarm rate on fire-free warehouse photographs (258 photographs frozen before scoring); neither involves a robot. Real photographs (P, P2) link the simulation to real appearance without a robot; they showed that the frozen threshold flagged 64/100 of fire-free photographs (hazy hillsides, sunsets) while the scores still rank well (AUROC 0.96).

## VII. Discussion and Limits

**Inputs, not just models, decide safety.** Trusting memory is efficient but its risk grows with how often the world changes, which the robot cannot observe without looking (H). Observation 1 and Proposition 2 settle whether checks happen: memory and any other advice decide only the order and strictness of checks, never their existence; the guarantee ends at the camera. **Moving first, then wording.** The largest single gain in $\varepsilon$ came from moving 0.4 m (G); in W's reference run with the v3.5 wording the look-around alone cut the misses from 144/150 to 48/150 (96 fewer), and the v3.6 wording and crops removed 45 more, so both contributed. **Value of pre-registration.** Of the registered claims, six failed (T1's 2% bound, F, T2, J, V4 (memory under random change) and W-b (hazard rule beats prior art)); the first two led to the look-around, and J kept us from adopting a second model. Without registration, each could have been rationalized away.

**Limits.** The guarantee is relative: G's plug-in $k\hat{\varepsilon} = 0.125$ for the hardest class (upper bound 0.57, 16 worlds) is superseded by W's estimate on 150 worlds; the order-independent $\varepsilon_{\text{any}} \le k\hat{\varepsilon}$ is the number to read, estimated at image level (W: 4.0% for $k = 2$, 7.8% for $k = 4$ with look-around; XD: *(pending)*). Exposure to a really burning station is not bounded. The price of advice repeated over many missions is bounded per account (an input channel): a credit that only drops advice limits the wasted visits and caution-caused refusals to a burst plus a recovery per mission (Prop. 3, in the supplement). Exposure is not part of this price, the model is descriptive, and its closed-loop test, study B, is pending. $\varepsilon_s$ differs by station and fire geometry, so the budget we measure on one class does not transfer to another. Scene text is perception and outside the guarantee, and so is the requested supply parsed from the operator's text (a text that flips it is not covered by Prop. 2). Study IF (72 flip and control texts) is registered to measure this; results pending. All results are from simulation, with one robot, two supply types, a stylized fire and sensor degradations rather than the appearance gap of real objects or smoke; the depot (study X, pending) is meant to narrow this gap but remains simulation (no grasping, real smoke dynamics or sensor noise). Campaign H scripts its changes (V tests random ones), and 8–16 pairs per closed-loop study are few. Barely visible fires (21–613 pixels) are excluded, and the guard is hand-written, not synthesized [20], [21]. Threats to validity: Supp. Sec. S9.

## VIII. Conclusion

A gate that rarely approves a wrong delivery needs a small error budget and a rule that nothing it did not perceive can exceed it. In one sentence: *Typing everything the robot did not perceive itself (memory, peer reports, the inventory, the operator's wording) as advice that may only reorder visits or tighten thresholds keeps the risk of an unsafe delivery, under any advice, even adversarial, within the check's own error budget* $\varepsilon_{\text{any}}$. The look-around shrank the budget (G: 11 to 1 of 16); in closed loop, verified memory kept a stale input from spending more than the budget (H: supported; V: model: supported; memory: not supported; pooled unsafe deliveries over 82 missions: M0 1, M2 1, naive memory 40) and tighten-only advice a hostile one (A: *(pending)*). The guarantee is relative and ends at the camera; scene text and realism are measured, not claimed. Everything is simulation, and a site-specific threshold is the first thing a real deployment must calibrate. All code, protocols and per-trial evidence are released.

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